COURSE-BASED PROJECT DOCUMENTATION | DOC-15

Comprehensive Testing Strategy

Verification Methodology, Test Harnesses, & Quality Assurance

1. Testing Philosophy & Multi-Tier Strategy

A rigorous multi-tier testing methodology was applied to ensure that grammar rules, lexical tracking, native bindings, and user interface interactions function flawlessly across all categories.

2. Testing Tiers

- ? Tier 1: Lexer Unit Testing ? Standalone token validation via lexer_driver.c and lexer_test.exe.
- ? Tier 2: Parser Batch Testing ? 59 automated test cases in tests/run_tests.py checking return codes and messages.
- ? Tier 3: Error Location Precision Testing ? Exact line validation after comments, blank lines, and text blocks.
- ? Tier 4: Stress & Boundary Testing ? Deeply nested record deconstructions, generic lambdas, and EOF boundaries.
- ? Tier 5: GUI Integration Testing ? 14 automated unit tests in tests/test_gui_backend.py testing UI widgets & FFI.

3. Automated Test Runner Architecture

The automated harness (tests/run_tests.py) iterates over test tuples (path, should_be_valid, description). It invokes check_syntax() via in-memory ctypes (or CLI fallback), validates (return_code == 0) == should_be_valid, and asserts that the error message contains the expected line number.